Autonomous Naval Epoxy-Polyamide Formulation Engine

Founder investment memo Customer: a top U.S. Navy prime

Bottom line: mvp20 is fundable as a deterministic evidence engine and CRO decision packet, not proven paint. Arithmetic is strong and reproducible; pot-life and corrosion have no authority; there is zero wet-lab data. The near-term spend is a small falsification test that decides whether the barrier index beats simple heuristics.

1. What is real / what failed

- **Real:** deterministic funnel (4320 enumerated → 3975 solved → 2444 hard-gate survivors); three tracks; source/claim registries; CRO recipe card; clean-slate provenance; run hashes.
- **Failed / bench-gated:** cure/pot-life model and corrosion model; both have no authority until measured.
- **Candidate Feasibility Band:** C01–C04 / C05,C06,C09,C10 / C07,C08 are bands, not evidence ranks; bench decides.
- **Blocked:** all supplier RFQ lines remain open for PFAS, TSCA, TAA/domestic, current SDS/TDS/CoA, lead time, MOQ, shelf/storage.

Track	Use	VS	VOC	WPG	PVC/Λ	R	ZnP	Status
Lineage-60	Public F160/F161 lineage reconstruction; authority benchmark, not an envelope-literal candidate	60.2%	285	9.24	8.2 / 0.15	0.78	no	major variance: public black/yellow solids are 60.2+/-2%, not 65+/-2%
Envelope-65	Envelope-literal 65% candidate; primary response to the customer envelope	65.0%	282	9.16	9.3 / 0.17	0.70	no	absent from the primary; color pigments are identification only
Enhanced-65	Optional process/corrosion-risk variant; functional zinc phosphate must earn its place at the bench	65.0%	282	9.35	10.9 / 0.20	0.70	yes, functional	functional optional inhibitor; not part of the primary non-functional-pigment response

2. Fundable vs not proven

Fundable now	Evidence discipline, arithmetic recompute, source/claim guardrails, CRO-screen-ready recipe packet pending declarations, and a falsifiable pairwise bet.
Not proven	Pot life, corrosion, adhesion, sprayability/sag, shelf life, procurement readiness, legal FTO, and any 65% qualification path.
What money buys	A controlled CRO test that either promotes the barrier hypothesis or kills it cheaply against solids/PVC heuristics.

Model	Purpose	Right to claim	Next promotion test
M2 VOC/solids/density/mix-r ...	Solve two-component recipes onto 1:1 ...	ARITHMETIC USABLE	ASTM by Method 24, D2697, D1475, drawdowns, ...
M12 Resin/curative property ...	Predict resin/curative equivalent ...	EXACT STRUCTURE ONLY	Train on estimated-tier Tg/viscosity models ...
M5 Known-material ...	Show the engine can rediscover known ...	SANITY CHECK	Run Lineage-60 and a commercial reference ...
M6 Solvency model	Score solvent blend compatibility ...	LIMITED SANITY	CRO compatibility, blush, phase-split, and ...
M7 Viscosity KU proxy	Band sprayability risk from solids ...	LIMITED SANITY	D562/D4400 spray/sag map and yellow D2805 ...
M8 Cure/pot-life model	Predict usable pot-life and cure ...	NO AUTHORITY	Week-1 KU-vs-time and usable-pot-life screen ...
M9 Corrosion rank model	Rank corrosion resistance from public ...	NO AUTHORITY	Side-by-side B117/D5894/D870/ISO 16773 EIS ...
M10 Barrier tortuosity model	Encode barrier design rules and lock ...	DIRECTIONAL ONLY	Register locked rank before panels; compare ...

M15 Bayesian CRO optimizer	Choose next experiments from measured ...	ROADMAP	Run first CRO batch, ingest raw outcomes, ...

3. Falsification test

The first test asks whether the engine beats simple heuristics, not whether the product is qualified. The key pairwise bet must hold; tie/noise means no promotion.

Locked order	Track	Barrier index	PVC %	ZnP
1	Lineage-60	0.477	8.2	no
2	Enhanced-65	0.474	10.9	yes
3	CRO-Robust-65	0.462	9.5	no
4	Envelope-65	0.460	9.3	no

Pairwise bet: the Enhanced-65 > Envelope-65 non-obvious call must hold in the measured data (the single most decisive, PVC-contrarian bet). Uncertainty: no calibrated uncertainty available. barrier-index deltas are too close to claim performance separation; the CRO pairwise result decides

Gate	Method	Target	Fail / inconclusive rule	Raw return field
gel_or_phase_split	visual check plus KU trend during induction and first ...	0 observed gel/phase-split events	repeat/review if control invalid; otherwise stop or reformulate on ...	visual_observation, ku_vs_time_3temps.raw_value, ...
pot_life_below_90_min	ASTM D562 KU-vs-time at 65/73/95F plus application ...	≥90 min at 73F	repeat/review on control failure or noisy tie; stop/reformulate if ...	pot_life_h.raw_value, ku_vs_time_3temps.raw_value, ...
voc_over_340_g_l	EPA Method 24 / ASTM D3960	≤340 g/L	inconclusive if worksheet missing; stop/reformulate if valid value ...	voc_g_l.raw_value, method24_worksheet_path
volume_solids_outside_63_67	ASTM D2697 or lab-accepted Method 24 solids worksheet	63-67%	inconclusive if raw replicates/worksheet missing; stop/reformulate if ...	volume_solids_pct.raw_value, replicate_values, ...

4. 30 / 60 / 90 day plan

Window	Work and gate
30	Supplier declarations/RFQs, 250 g mixed screen, 1 kg/component bench batch; KU-vs-time, compatibility, density, Method 24, D2697, drawdowns. Advance only if pot-life, compatibility, and arithmetic close.
60	Side-by-side corrosion/adhesion battery on Lineage-60, Envelope-65, Enhanced-65, and commercial reference if obtainable. Advance only if controls hold and the decisive pair is not tied.
90	5 gal kit only if gates pass; ingest raw outcomes, update model rights, close declarations, choose Lineage-60 / 65% / neither.

5. Moat: current vs future

Current moat	Future moat
Deterministic constraint solving; audited source/claim discipline; reproducible hashes; CRO data schema; falsifiable model-rights discipline.	Proprietary CRO data flywheel, process/QC data, active-learning optimizer, standards RAG, patent claim graph, trained corrosion/cure/rheology models.

6. Run receipt and provenance

Audit artifact	Status	Detail
Whole-run output hash (sha256)	PRESENT	e15c20df360d12cd18e42888...
Code tree hash (sha256)	PRESENT	31b73a3b148091cc28c77226...
Git commit	UNAVAILABLE	Command '['git', 'rev-parse', 'H...
Frozen scoring weights	PRESENT	published in funnel + manifest
Gate/scoring config versions	PRESENT	mvp20.gates.v1+cro_capture.v2 / mvp20.funnel_weights.v1
MC seed / source-query hash	PRESENT	2001 / 74509b975ab9293e...
Barrier locked-rank hash	RECOMPUTES	197cf16aa46bca218a53...
Input-PDF sha256 (x4)	PRESENT	government slash sheets
No manual candidate edits after run start	ATTESTED	deterministic enumeration; byte-identical reruns
Third-party pre-registration	REQUIRED	before panels are cut

Item	User	Engine	Gen	Validation	Guard
Human technical envelope / problem constraints	Y	N	Y	N	N
F160/F161 public MIL formula lineage	N	Y	N	Y	Y
F150/NCP Type IV partial-TDS anchors	N	Y	N	Y	N
Evonik SPF A702B75P1 starting-point formulation	N	Y	N	Y	N
Supplier raw-material families	N	Y	Y	N	N
Corrosion literature	N	Y	N	N	N
QPL/F-718 authority records	N	Y	N	N	Y

Autonomy boundary: What "autonomous" means here: human-defined problem envelope; machine-executed source retrieval, constraint solving, gating, scoring, and CRO-test design; human-controlled lab / EHS / legal / procurement. It is deterministic machine-executed search + audit within a human-set envelope — not autonomous science.

7. Hard gaps

Gap	Severity	Next action	Kill criterion
Corrosion	HIGHEST	B117/D5894/D870/ISO 16773 EIS	controls fail, nulls win, or decisive pair tied
Pot life	HIGH	KU-vs-time at 65/73/95F	<90 min usable or phase split/gel
Supplier declarations	HIGH	PFAS/TSCA/TAA/SDS/TDS/CoA RFQs	cannot source declaration-compliant BOM
FTO	MED	counsel review	no written freedom-to-operate path

KPI	Current value
Constants total	83
Two-source constants	11
Arithmetic-crosschecked constants	8
Single-source constants	48
Manual-required value comparisons	49
Load-bearing violations	0
Confidence	arithmetic HIGH; source values MEDIUM; sourcing LOW

8. Recipe readiness

Variant	Part	Role	g basis	Lines
black	part_A	epoxy side	137.4	dgeba_828, silica, dispersant, defoamer, thixotrope, iron_oxide_black, naphtha
black	part_B	polyamide-curative side	102.2	ancamide_350a, silica, dispersant, thixotrope, n_butanol
yellow	part_A	epoxy side	136.8	dgeba_828, silica, dispersant, defoamer, thixotrope, iron_oxide_yellow, naphtha
yellow	part_B	polyamide-curative side	102.2	ancamide_350a, silica, dispersant, thixotrope, n_butanol

9. Known-control checks

Display	Check	P/F	Evidence	Validates	Does not validate
1	F160 black public formula	PASS	KNOWN FULL-BOM	formula parsing; density/VOC envelope arithmetic; known-control sanity ...	top recovery proof; candidate perfor mance; corrosion performance; pot ..
2	F161 yellow public formula	PASS	KNOWN FULL-BOM	formula parsing; density/VOC envelope arithmetic; known-control sanity ...	top recovery proof; candidate perfor mance; corrosion performance; pot ..
3	Evonik SPF A702B75P1	PASS	KNOWN FULL-BOM	formula parsing; density/VOC envelope arithmetic; known-control sanity ...	top recovery proof; candidate perfor mance; corrosion performance; pot ..
4	NCP MIL-DTL-24441 Type IV anchor	PASS	KNOWN PARTIAL-TDS	class-envelope recovery; 1:1 epoxy-polyamide precedent check	full-BOM reconstruction; corrosion performance; pot life; ...
5	International Formula 150 anchor	PASS	KNOWN PARTIAL-TDS	class-envelope recovery; 1:1 epoxy-polyamide precedent check	full-BOM reconstruction; corrosion performance; pot life; ...
6	over-pigmented (talc 300 phr)	PASS	NEGATIVE CONTROL	fail-closed gate behavior; negative-control rejection	candidate performance; candidate cor rosion; candidate pot life; ...
7	stoichiometry R=1.3	PASS	NEGATIVE CONTROL	fail-closed gate behavior; negative-control rejection	candidate performance; candidate cor rosion; candidate pot life; ...
8	80% volume-solids demand	PASS	NEGATIVE CONTROL	fail-closed gate behavior; negative-control rejection	candidate performance; candidate cor rosion; candidate pot life; ...

10. Sourcing blockers

Blocker	Status
RFQ lines	open
PFAS	unknown / RFQ required
TSCA	unknown / RFQ required
TAA/domestic	unknown / RFQ required
Current SDS/TDS/CoA	unknown / RFQ required
Lead time/MOQ/shelf/storage	unknown / RFQ required

11. Decision outcomes

{"controls_fail": "repeat/review; never treat candidate data as pass", "envelope65_beats_enhanced65": "demote zinc-phosphate hypothesis", "lineage_60_wins_all_65_fail": "do not qualify 65%; use Lineage-60 path if otherwise obtainable", "noisy_or_tied_decisive_pair": "no model promotion; collect more data or reformulate"}

Appendices

Dense evidence is in TECHNICAL_APPENDIX.pdf. Molecule property/generator details remain in MOLECULE_ENGINE.pdf.